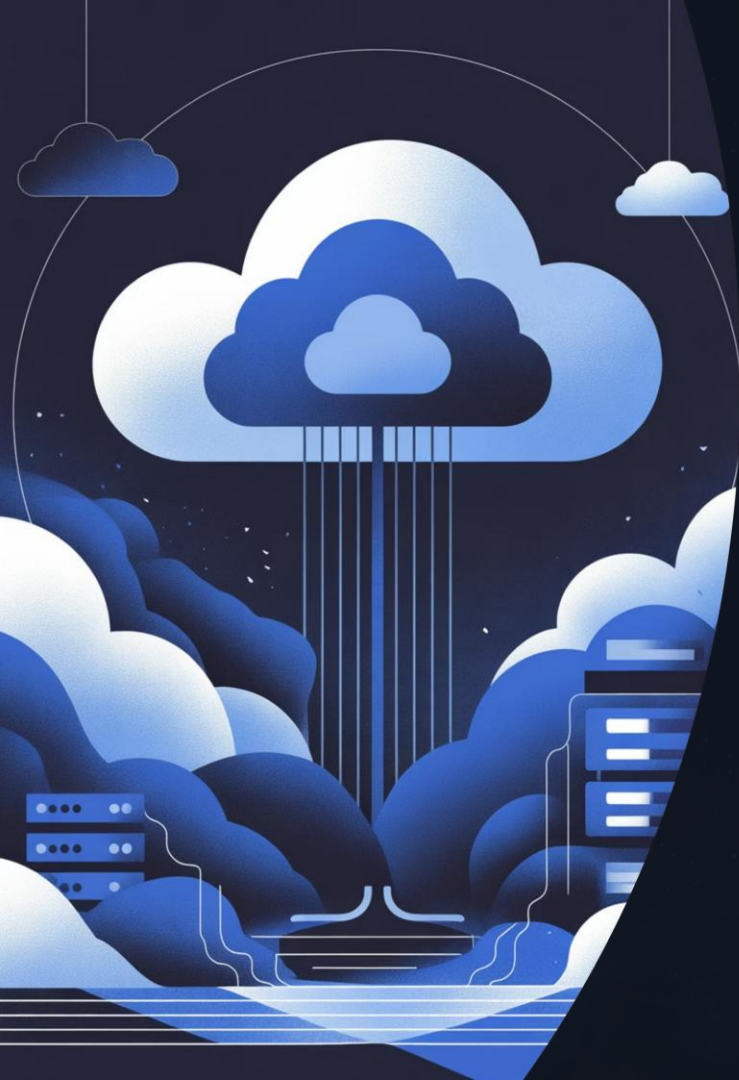




AI-Powered Application Design and Planning

Use Claude Code to Design and Plan Implementations

[Derek Ashmore](#)

**AI
Summit**

Brussels · May 26, 2026

Meet Derek Ashmore

A Professional Geek since 1987, I bring decades of experience to the evolving landscape of IT.

My Expertise Includes:

- Applying AI tooling for IT innovation
- Driving Application Transformation initiatives
- Pioneering Infrastructure Automation solutions

Yes — I still code!



MASTERING THE SHIFT

Collaborating with Teams
of AI Coding Agents

Derek C. Ashmore

Agenda

A roadmap of our discussion on AI-powered application design and planning using Claude Code.

1 AI Agents for Architecture & Planning

How Claude Code and RuFlo accelerate non-coding work — from 12-Factor analysis to implementation planning — faster and more completely than human teams.

2 Real-World Case Studies

Practical examples from the field: modernizing the Orienteer CRM, SaaS migration analysis, GitHub Runner design, security compliance, and more — with honest answers on trusting AI output.

Scan to follow along



Real-Life Stories from the Field

Real-world examples of Claude Code helping teams accelerate non-coding work across architecture, planning, compliance, and modernization.

Application Modernization

- 12-Factor analysis on an unknown codebase
- Implementation planning to remediate the unknown codebase

New Amazon/Walmart Item List Management Application

- Architecture
- Design
- Cost estimate
- Implementation planning

GitHub Runner Implementation

- Design and plan in-network runner hosting with cloud privileges to support IaC

Security Compliance

- Architecture
- Design
- Cost estimate
- Implementation planning

New Application to Support Split-Brain DNS Entries for a Large Enterprise

- Architecture
- Design
- Cost estimate
- Implementation planning

Many More...

The possibilities are expanding rapidly.



Why RuFlo?

Quick Setup

Install and run within minutes, not hours or days. Default configuration works for the vast majority—it just works.

Powerful Resources

Dozens of MCP tools and 16 specialized agents (coder, researcher, analyst, tester, coordinator, etc.).

Smart Assembly

RuFlo assembles the agent team based on your request, determining the right mix of specialists.

- ❑ Claude Code also offers an experimental Teams feature — a direct alternative to RuFlo. [See my real-world bake-off comparison.](#)

Why Architects Need AI Agents Too

Not just for coders — AI agents transform how architects work.

4h

Orienteer analysis

12-Factor analysis, remediation plan, and SaaS replacement analysis completed in about 4 hours total.

Days

Human team timeline

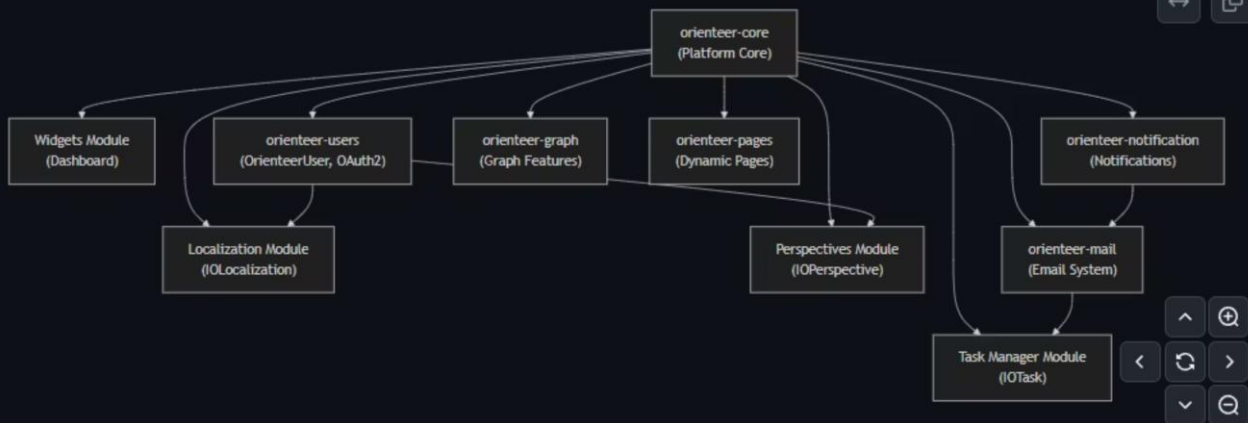
The same work would have taken days or weeks for a human team.

Hours

Documentation speed

Complete documentation is written in hours, not days.

Module Dependencies



AI Agents

Improve Quality

Designs are more complete — and Claude fact-checks too.

- More thorough than manual designs — AI has time to be complete
- Catches missed details: encryption, monitoring, log retention
- Claude fact-checks its own output
- Judge for yourself: [Orienteer Executive Summary](#)



12-Factor Compliance Scorecard

Factor	Score	Status	Critical Issues
I. Codebase	7.5/10	✓ Good	Single repo, multiple deployable artifacts
II. Dependencies	8.5/10	✓ Good	Explicit Maven dependencies, no vendoring
III. Config	3/10	● Critical	Hardcoded credentials, secrets in codebase
IV. Backing Services	6/10	● Moderate	Embedded database default, limited abstraction
V. Build, Release, Run	7/10	✓ Good	Docker support, but config in build
VI. Processes	2/10	● Critical	Stateful sessions, requires sticky routing
VII. Port Binding	10/10	✓ Excellent	Self-contained with embedded Jetty
VIII. Concurrency	4/10	● Poor	Monolithic, no process separation
IX. Disposability	3/10	● Poor	No graceful shutdown, slow startup
X. Dev/Prod Parity	4/10	● Poor	Environment-specific code paths
XI. Logs	6/10	● Moderate	Stream-based but basic implementation
XII. Admin Processes	7/10	✓ Good	Robust migration system, console support

AI Agents Provide Tactical Advantages

A forcing function for clarity, accuracy, and better decisions.

- Forces a central source of truth — bad requirements produce bad output
- Forces clarity — vague or conflicting requirements cause unexpected results
- Requirements documented in markdown for easy revision ([Example requirements](#))
- Enables quicker conversations with management and product owners
- Agent teams run alternate scenarios quickly and with high quality
- Enables better decisions

Application requirements

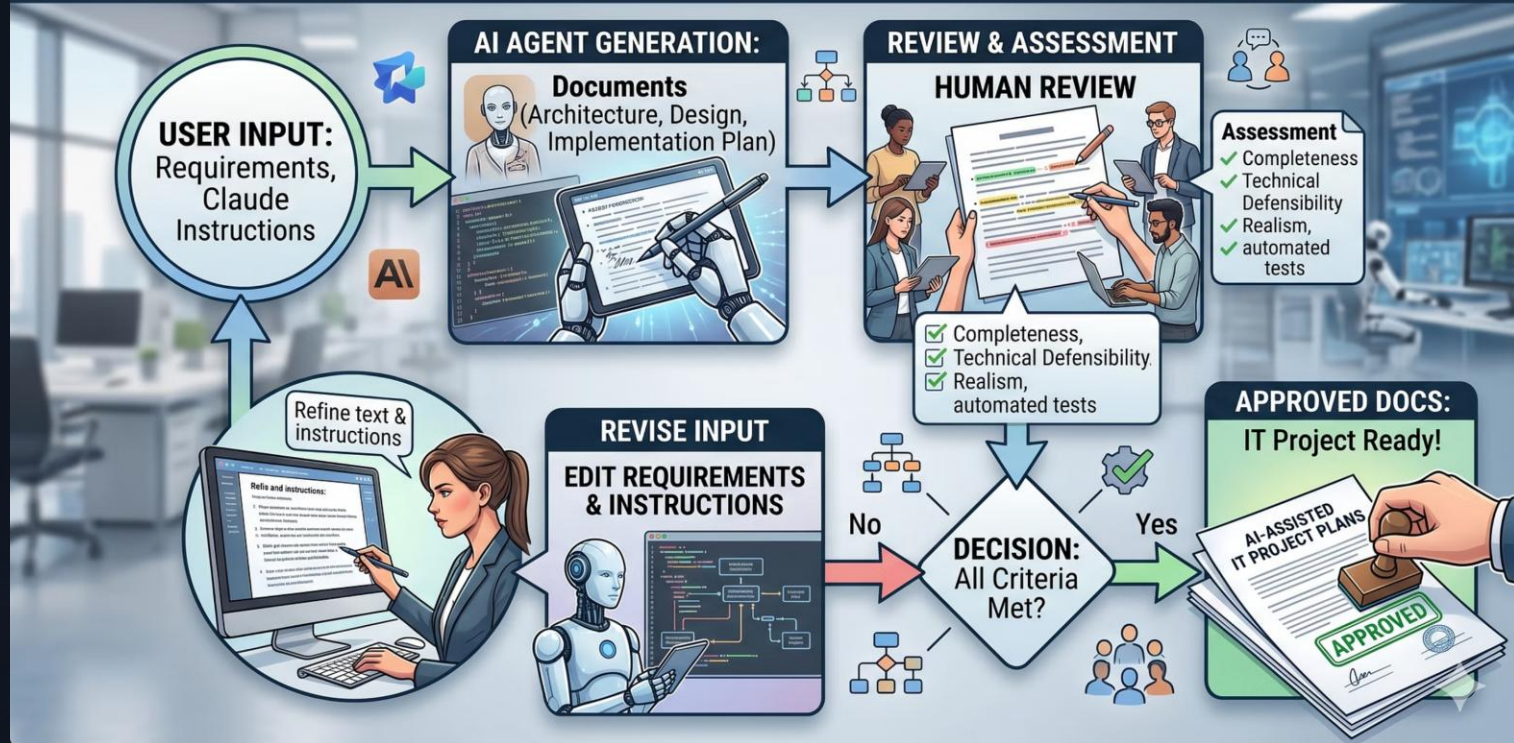
- The application that will analyze historical daily spot price data using [technical analysis](#) techniques.
- The assets classes considered will be equities. No forex, commodities, cryptocurrencies, or anything else.
- Consider long trading positions only. No short selling recommendations.
- All long positions will use the entire balance of the trading account for buys and sells. Long positions are either completely in the market or out of it.
- The application will analyze one asset at a time.
- The application will report investment strategies that best fit the asset over the length of the analysis. An investment strategy is defined as one or more technical analysis indicators that are used to signal a buy or sell. Examples of technical indicators are moving averages, RSI, MACD. Best fit means indicates buys and sells most profitably over the entire analysis.
- The application will provide a verbose mode that will document what technical analysis indicators it examined and rejected. Include a summary of what was found.
- Make indicators that produce no trades available in verbose output.
- The application will be run once a day to accommodate new daily price data.
- The application should benchmark indicator performance against buy-and-hold and the S&P 500 index.

Trade profitability reporting

- The application will report for indicators chosen trade-by-trade entry/exit dates with trade profit or loss.
- The application will report indicated buys and sells and the profitability for each trade.
- All trades will be complete. That is, buys will use all available capital and sells will include all shares.
- The application will provide summary statistics over time such as average length of trade, average profitability, average length of time between trades.
- The application will allow the user to indicate the initial trading account size and starting point for trade profitability reporting. Default to account size \$10,000 and starting point January 1, 2000.
- The application is stateless and will not require persistence between runs. I understand that results may change when new data is analyzed.
- The application will provide output that includes an actionable signal ("BUY today" / "HOLD" / "SELL today").

Agent-Based Design and Planning Workflow

ITERATIVE DESIGN-REVIEW-REVISE LOOP: AI AGENT FOR IT DOCUMENTS



Implementation Planning: Specification Statements

To guide RuFlo in generating a comprehensive implementation plan for Orienteer's 12-Factor compliance, the following [instructions](#) were provided:

1. **Leverage Analysis** — Use the 12-factor folder analysis as foundational input.
2. **Agentic Execution** — Implementation performed by agentic engineers via Claude-Flow and Claude Code.
3. **Directed Output** — Deposit the generated plan into the 12-factor-plans folder.
4. **No Immediate Changes** — Do not modify Orienteer during the planning phase.

Executive Summary

Total Investment Required

Category	Estimate	Details
Human Labor	\$800K-1.2M	5-8 FTE for 8-10 months + part-time support
Infrastructure	\$40K-60K	Cloud resources, testing tools, CI/CD
Training & Tools	\$15K-25K	Licenses, courses, certifications
Contingency (15%)	\$130K-190K	Risk buffer
Total Investment	\$985K-1.475M	Complete transformation

ROI Analysis

Benefits (Annual):

- Infrastructure cost reduction: \$150K-250K/year (30-50% savings)
- Reduced maintenance: \$100K-150K/year (fewer production incidents)
- Faster feature delivery: \$75K-100K/year (2x deployment frequency)
- **Total Annual Benefit: \$325K-500K/year**

Payback Period: 24-36 months

SaaS Migration Analysis Process

- Delivered results in ~15 minutes vs weeks for human teams
- Key insights:
 - Would benefit from specific client meta-data
 - Start of a conversation, not the end

Top 3 Recommended SaaS Alternatives

Primary Recommendation: Salesforce Platform

Overall Score: 9.2/10

Key Strengths:

- **Enterprise-Grade:** Industry-leading security, compliance, and reliability
- **Comprehensive Feature Set:** 95% capability coverage of Orienteer requirements
- **Ecosystem Maturity:** Extensive marketplace, integrations, and developer community
- **Proven Migration Path:** Well-established migration methodologies and tooling

Migration Complexity: Medium (3-6 months) **Annual Cost Range:** \$150,000 - \$400,000 for typical enterprise deployment

Best Fit For: Large enterprises requiring maximum feature coverage and ecosystem integration

Alternative Recommendation: Microsoft Power Platform

Overall Score: 8.7/10

Can I Trust AI Analysis?

Reframe the question

Do I trust Claude Code more or less than a human team?



All analysis have error margins

Whether from AI or humans, no plan is perfect.



Claude ignores corporate politics

Objective analysis without organizational bias.



Unbeatable speed

15 minutes from start to finish, with rapid follow-up responses.

Key Takeaways



For Architects

Agentic engineering prepares you for modernization conversations with more complete, consistent analysis than human teams provide. Run alternative scenarios quickly for management.



For Team Leads

Prepare for migrations more completely and quickly than manual approaches. AI sometimes surfaces issues humans forget.



For Agile Teams

Formulate larger-scope stories for agentic engineers. Whether migration or upgrade, AI helps break down work effectively.

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Thank You!

We appreciate your time and interest in AI-powered application modernization.

For further inquiries or to discuss how Claude-Flow can transform your applications, please feel free to reach out:

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- derek.ashmore@asperitas.consulting

